Bias vs Variance in Machine Learning

1. T-shirt Analogy

- An overfit t-shirt is too tight, so it'll not fit if your body changes, just like an overfit model is too sensitive to training data.
- An underfit t-shirt is too loose & does not fit well, similar to an underfit model that cannot capture the underlying pattern in the data.
- A perfect-fit t-shirt fits well even with slight changes in body size, just like a balanced model with low bias & low variance.

2. Housing Price Example (1 feature:- Square footage)

- In the case of overfitting, the model is too complex & fits the training data exactly, resulting in a training error close to zero but a high test error that varies significantly depending on which training samples are used.
- In the case of underfitting, the model is too simple, & cannot capture the data pattern, resulting in high training error, & the test error is similar to the training error, meaning it does not vary much with different training sets.

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- A balanced model captures the pattern in the training data well, resulting in low training error & low test error, & it generalizes consistently across different training samples

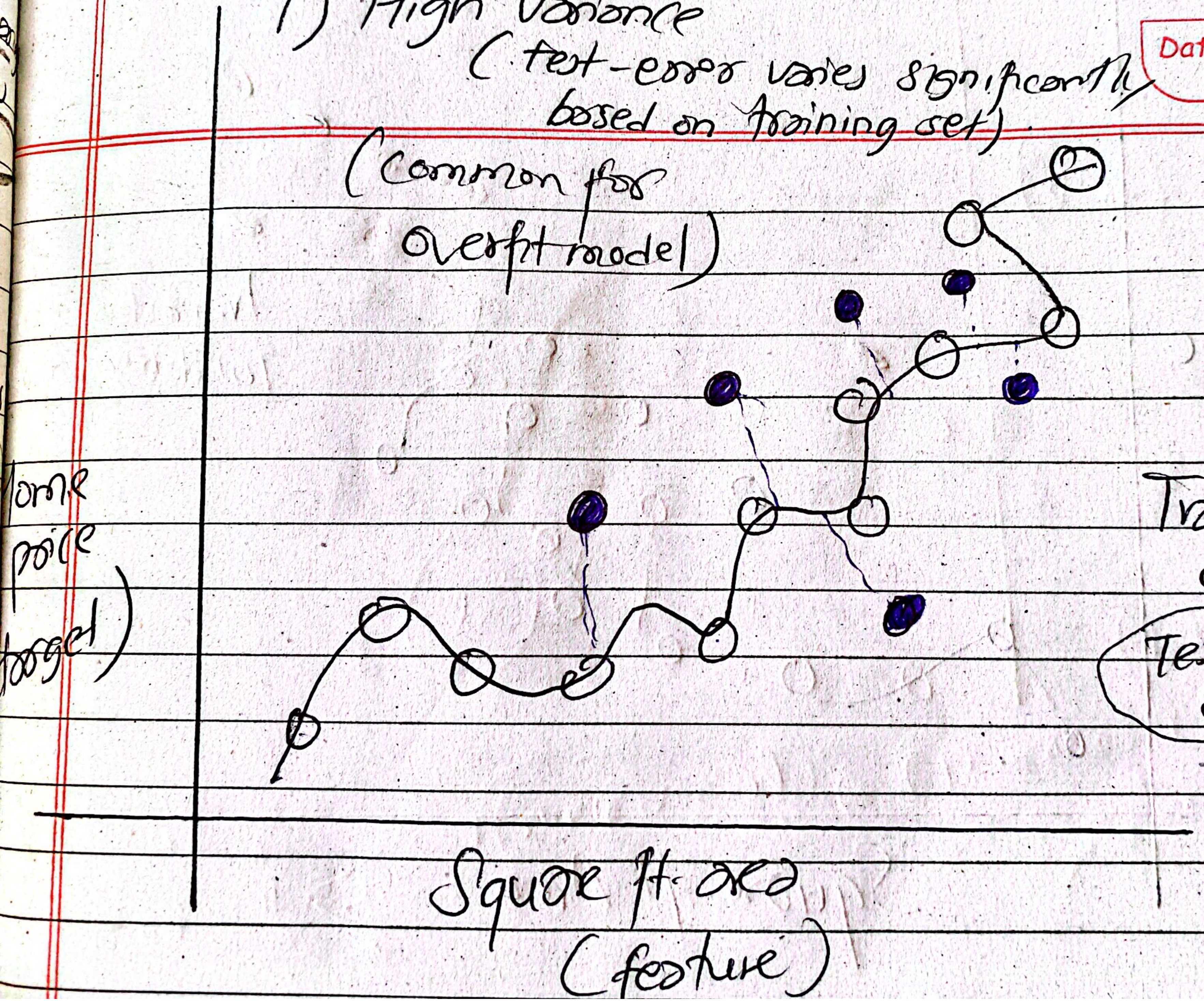
3. Understanding Variance

- High variance occurs when the test error [illegible] significantly depending on which training samples are selected, which is common in overfit models.
- Low variance occurs when the test error remains stable across different training sets, which is seen in underfit & balanced model
- Variance is always measured on the test data not the training data.

4. Understanding Bias

- High Bias occurs when the model cannot accurately capture the training data pattern, leading to high training error, which is typical in underfit models.
- Low bias occurs when the model fits the training data accurately, resulting in low training error, which is seen in overfit & balanced models.
- Bias is always measured on the training data, not the test data.

1) High Variance
(test-error varies significantly based on training set)

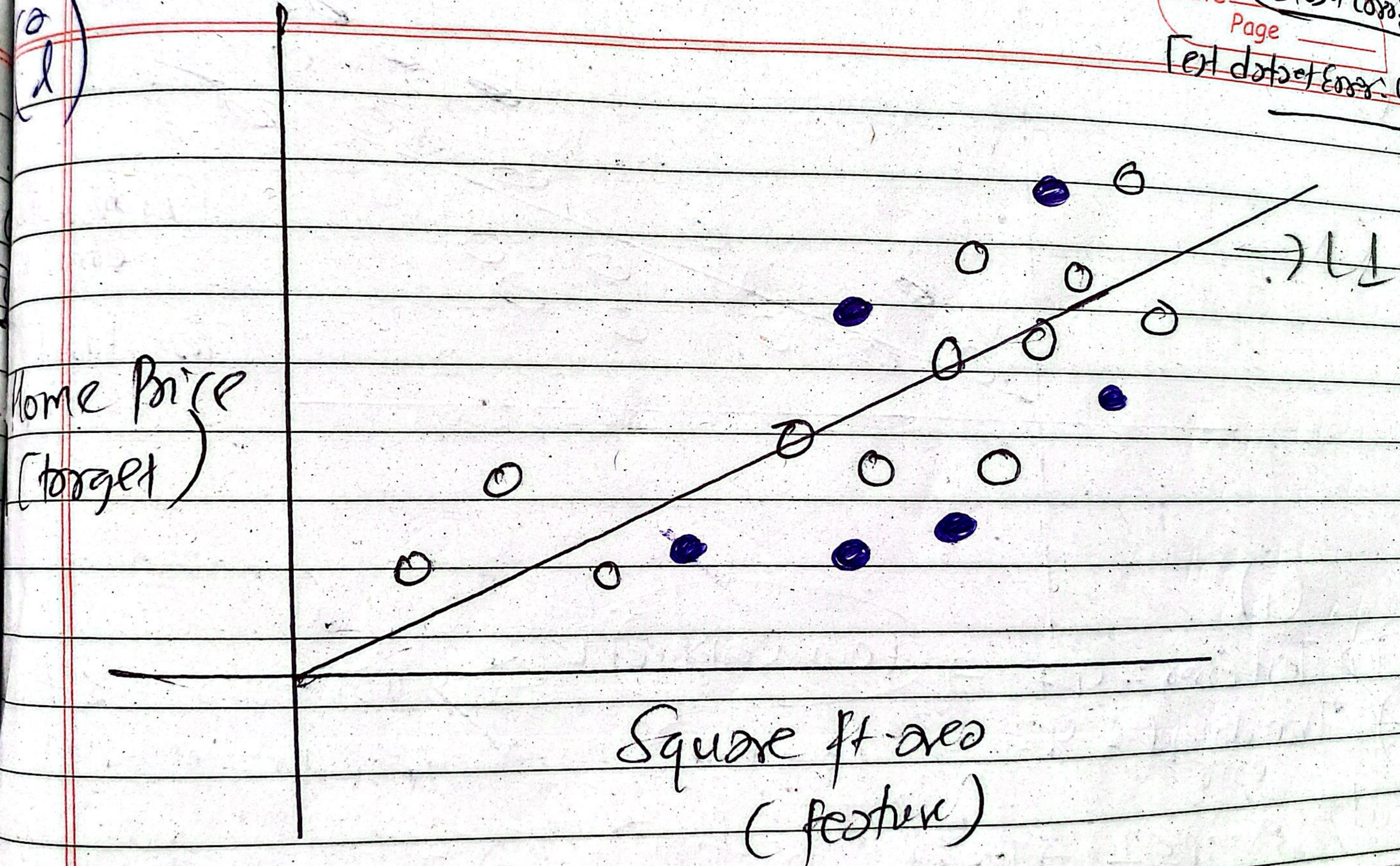


Train dataset
Error: 0

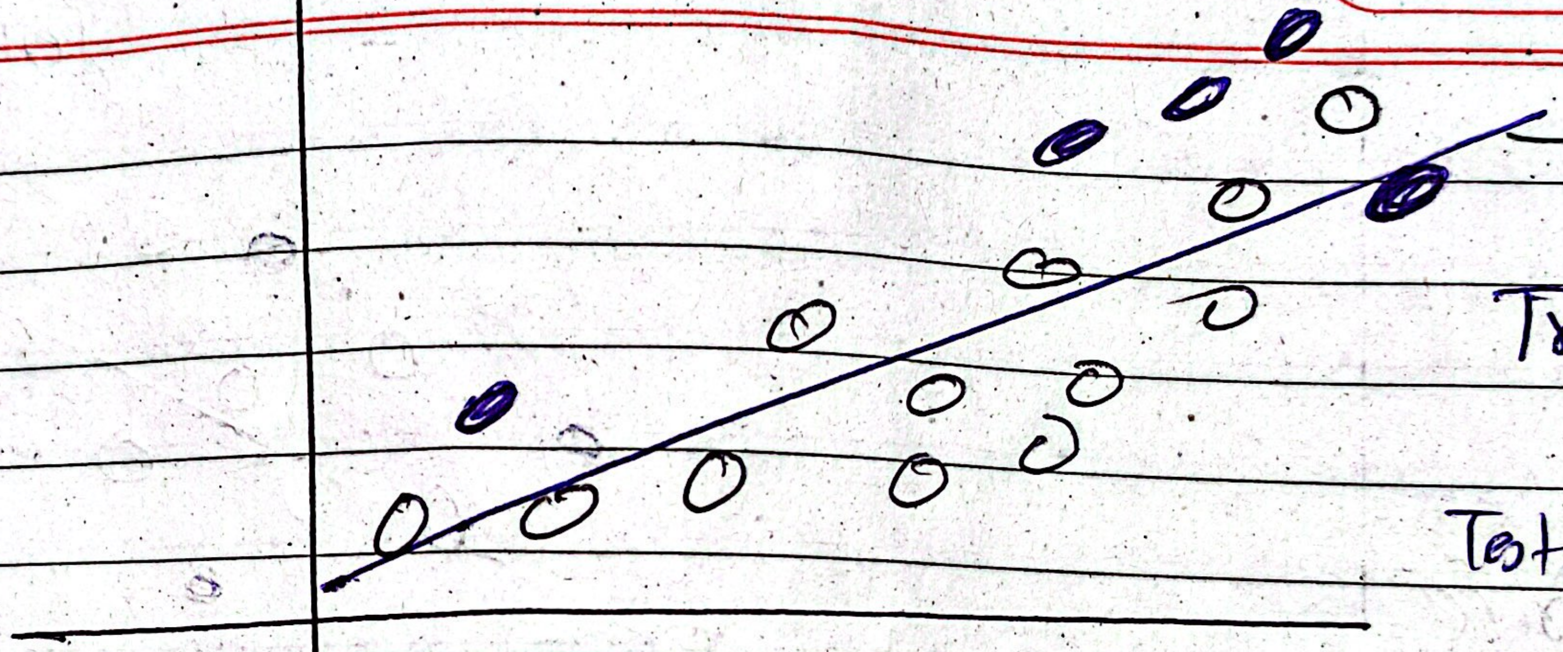
Test dataset
Error: 27

Train dataset Error: 43

Test dataset Error: 47



(ii)



→ L2

Train dataset error: 41

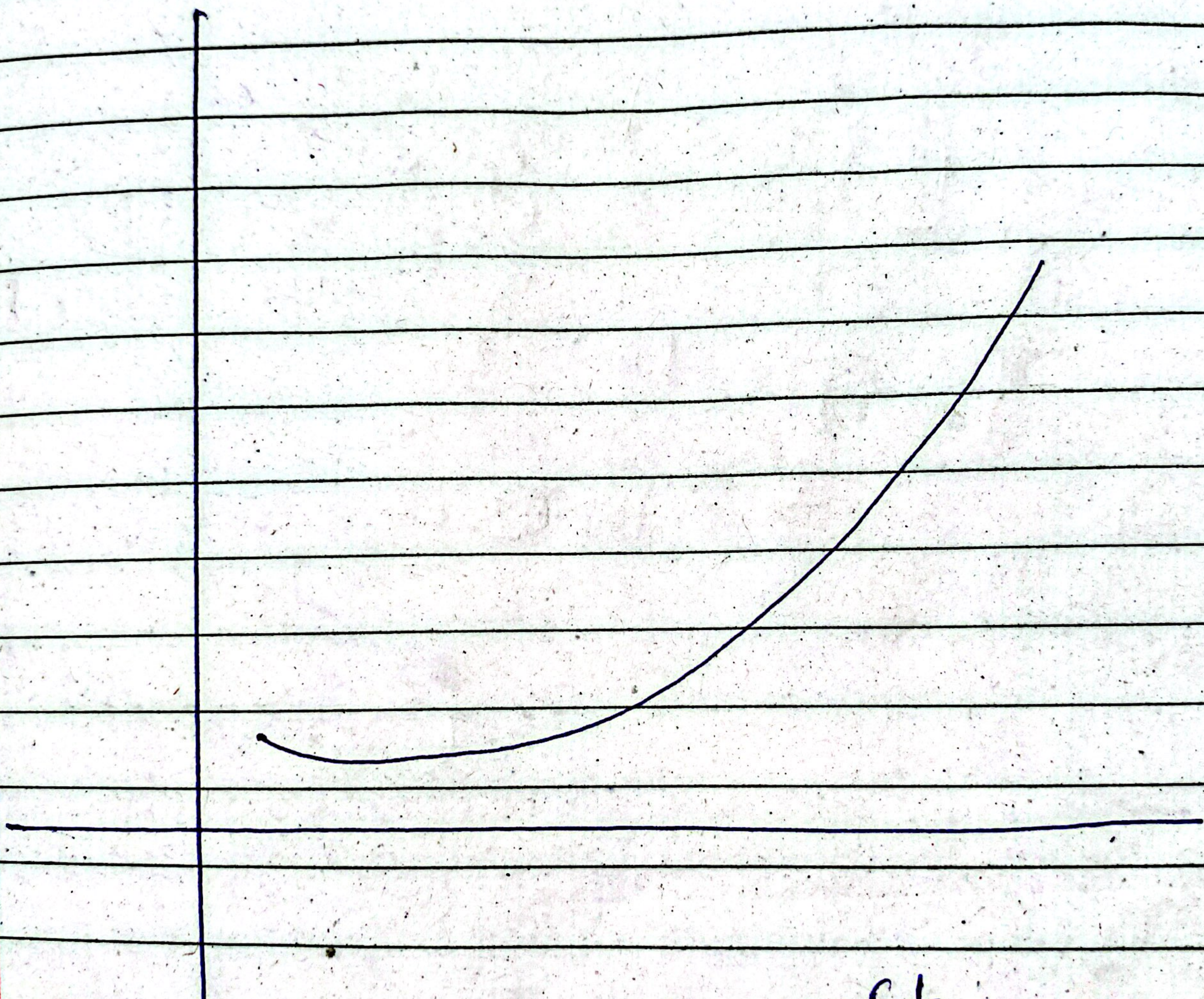
Test dataset Error: 37

(I) (II)

⇒ Test Error = 47 → low variance ← ⇒ Test Error = 37

⇒ Train dataset error = 43 → low Bias ← → Train dataset error = 41

But, Example 1 → Train dataset error = 0 High Bias ⇒ Train dataset error = 41



Balanced fit

~~Bias~~ low → more close to center (true value)
(training error ↓)

Variance low
→ ~~more~~ close they are together
(testing error ↓)

Practically

Alphabet
1. Bias → Trains
2. Variance → Testing

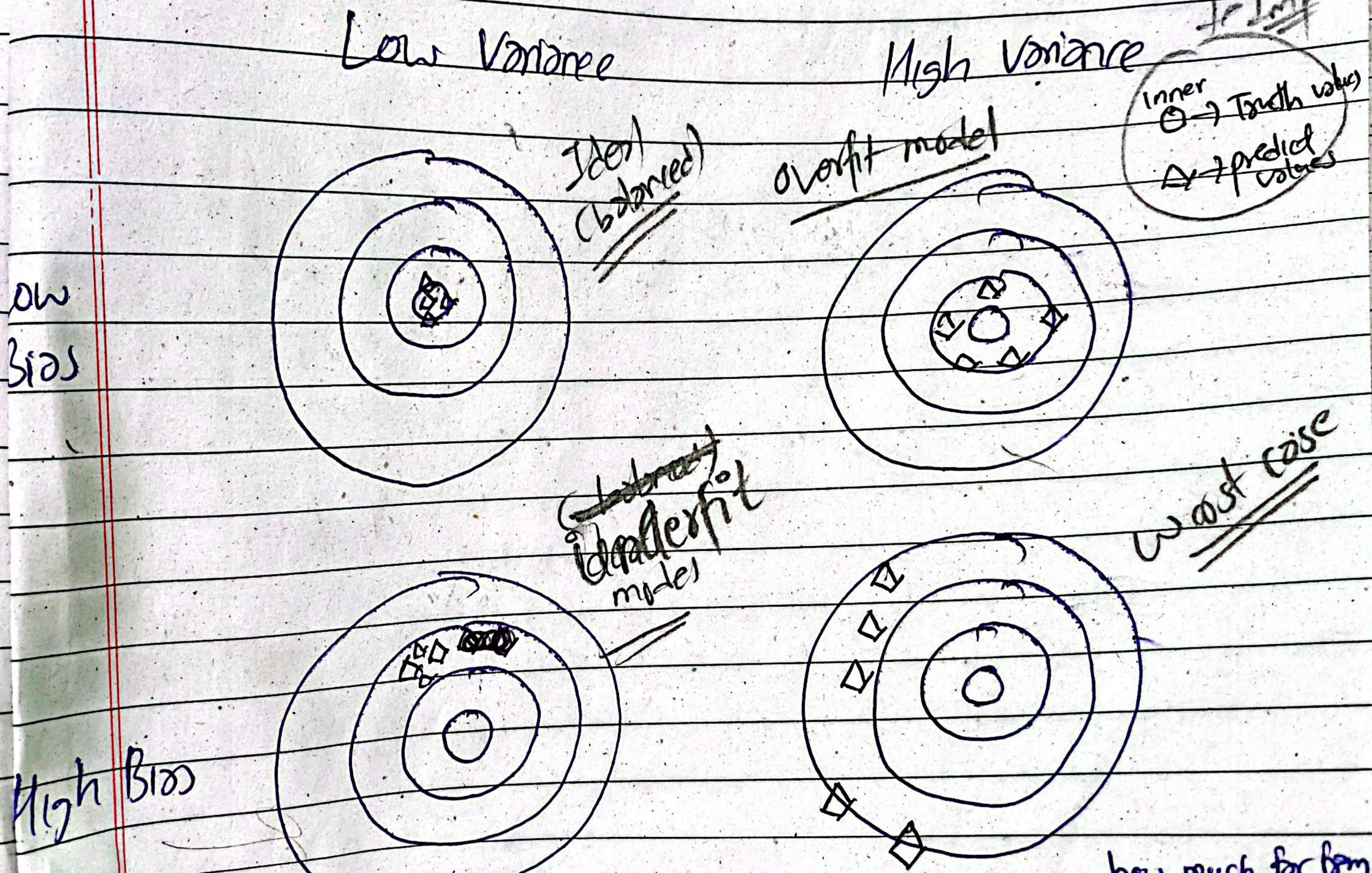
Bull's Eye Analogy for Bias & Variance.

- The inner circle represents the true values & the white diamond represent the predicted values.
- low bias & low variance occur when the predictions are close to the center & close together, which represents the ideal scenario.
- low bias & high variance occur when predictions are near the center but scattered, which represents an overfit model.
- High bias & low variance occur when predictions are far from the center but clustered together, representing an underfit model.
- High bias & high variance occur when predictions are far from the center & scattered, which is the worst-case scenario

Relationship Between Model Complexity & Error

A simple model that is underfit has high training error, high test error, high bias, & low variance.
A complex model that is overfit has low training error, high test error low bias, & high variance
A balanced model has low training error, low test error, low bias, & low variance, making it ideal for generalization.

Low variance occurs when the predi...



how much far from center → Bias
how close together → variance

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7. Techniques to achieve a Balance fit

- Cross-validation helps in evaluation model performance across multiple train-test splits to avoid overfitting.

  ([illegible])

- Regularization methods like L1 & L2 penalize overly complex models to reduce variance.
- Dimensionality reduction removes irrelevant or noisy features, helping the model generalize better.
- Blagging & boosting combine multiple models to reduce variance & improve predictive performance. Ensemble Technique

8. Key Takeaways

training error ↓ testing error ↑

- Overfit models have low bias but high variance, making them sensitive to the choice of training data.

  training error ↑

- Underfit models have high bias but low variance, meaning they cannot capture the data pattern accurately.
- Balanced models have low bias & low variance, generalizing well to new, unseen data.
- The goal in machine learning is always to achieve model with low bias & low variance for accurate predictions.

Model Complexity	Training Error	Test Error	Bias	Variance	Example
1. Simple (underfit)	High	High	High	Low	Linear regression on complex data
2. Complex (overfit)	Low	High	Low	High	Polynomial regression fitting all points
3. Balanced	Low	Low	Low	Low	Well-regulated model

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